Closed-Form Likelihood Computation for Birth–Death Phylogenetic Models

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Abstract

We derive closed-form solutions for the likelihood of reconstructed phylogenetic trees under the constant-rate birth–death (CRBD) and time-dependent birth–death with constant turnover (TDBD) models. The key observation is that when the extinction-to-speciation ratio $\varepsilon = \mu/\lambda$ is constant, the Riccati ODE for the extinction probability $E(\tau)$ becomes separable and admits an explicit solution. The branch propagation factor for the density $D(\tau)$ also has a closed form obtained by substitution. For the multi-state BAMM model, we exploit the upper-triangular structure of the transition matrix to solve absorbing (foreground) states analytically, reducing the ODE dimension for coupled (background) states. All formulas are verified against high-precision numerical ODE solutions (118 tests, tolerance 10^{-6}).

Contents

1 Notation and Setup

Consider a reconstructed (extant-only) phylogenetic tree with n tips, $n-1$ internal nodes, and $2n-2$ branches. We use the *age* convention:

- τ = time before present (age): present = 0, root = T.
- Each branch connects a child node at age τ_c to a parent node at age $\tau_p > \tau_c$.
- All leaves have age $\tau_{\text{leaf}} = 0$.

Two key quantities propagate through the tree:

- $E(\tau)$: probability that a lineage existing at age τ goes completely extinct before the present.
- $D(\tau)$: probability density of the observed subtree rooted at age τ.

The full log-likelihood of the tree under any birth–death model is:

$$\log \mathcal{L} = \log \Gamma(n) + \log D_{\text{root}} - 2 \log\bigl(1 - E(T)\bigr), \tag{1}$$

where $\Gamma(n) = (n-1)!$ accounts for label permutations and the last term conditions on the root's two initial lineages both surviving to the present.

2 The Riccati Structure

For a single-state birth–death process with speciation rate $\lambda(\tau)$ and extinction rate $\mu(\tau)$, the extinction probability satisfies:

$$\frac{\mathrm{d}E}{\mathrm{d}\tau} = \mu(\tau) - \bigl(\lambda(\tau) + \mu(\tau)\bigr) E + \lambda(\tau) E^2, \qquad E(0) = 0. \tag{2}$$

This is a **Riccati equation**. In general, Riccati equations do not have closed-form solutions. However, when $\mu(\tau)/\lambda(\tau) = \varepsilon$ is constant, (2) factors as:

$$\frac{\mathrm{d}E}{\mathrm{d}\tau} = \lambda(\tau)\,(E-1)(E-\varepsilon). \tag{3}$$

This is **separable**: the left-hand side depends only on E, and the right-hand side's τ-dependence enters solely through $\lambda(\tau)$.

The density $D(\tau)$ satisfies a linear ODE along each branch:

$$\frac{\mathrm{d}D}{\mathrm{d}\tau} = -\bigl(\lambda(\tau) + \mu(\tau) - 2\lambda(\tau)\,E(\tau)\bigr) D. \tag{4}$$

Once $E(\tau)$ is known analytically, this can be integrated directly.

3 TDBD Closed-Form Solution (General Case)

3.1 Model Definition

The TDBD model has time-varying rates with constant turnover ratio:

$$\lambda(\tau) = \lambda_0\, \mathrm{e}^{z(T-\tau)}, \qquad \mu(\tau) = \varepsilon\, \lambda(\tau), \qquad 0 \le \varepsilon < 1.$$

Define the cumulative speciation rate:

$$\Lambda(\tau) \;=\; \int_0^\tau \lambda(s)\,\mathrm{d}s \;=\; \begin{cases} \dfrac{\lambda_0}{z}\big(\mathrm{e}^{zT} - \mathrm{e}^{z(T-\tau)}\big), & z \neq 0, \\ \lambda_0\,\tau, & z = 0. \end{cases} \tag{5}$$

3.2 Extinction Probability $E(\tau)$

Proposition 1 (TDBD extinction)**.** *The solution to* (3) *with* $E(0) = 0$ *is:*

$$\boxed{E(\tau) = \frac{\varepsilon\,(1-\psi)}{\varepsilon - \psi}}\,, \qquad \psi(\tau) = \exp\!\big((1-\varepsilon)\,\Lambda(\tau)\big). \tag{6}$$

Proof. Separate variables in (3):

$$\frac{\mathrm{d}E}{(E-1)(E-\varepsilon)} = \lambda(\tau)\,\mathrm{d}\tau.$$

Partial-fraction decomposition ($\varepsilon \neq 1$):

$$\frac{1}{(E-1)(E-\varepsilon)} = \frac{1}{1-\varepsilon}\left(\frac{1}{E-1} - \frac{1}{E-\varepsilon}\right).$$

Integrating both sides from ($E = 0$, $\tau = 0$) to (E, τ):

$$\frac{1}{1-\varepsilon}\,\ln\left|\frac{E-1}{E-\varepsilon}\right| - \frac{1}{1-\varepsilon}\,\ln\frac{1}{\varepsilon} = \Lambda(\tau).$$

Rearranging:

$$\frac{E-1}{E-\varepsilon} = \frac{1}{\varepsilon}\,\mathrm{e}^{(1-\varepsilon)\Lambda(\tau)} = \frac{\psi}{\varepsilon}.$$

Solving for E:

$$E - 1 = \frac{\psi}{\varepsilon}(E-\varepsilon) \implies E\left(1 - \frac{\psi}{\varepsilon}\right) = 1 - \psi \implies E = \frac{\varepsilon(1-\psi)}{\varepsilon - \psi}.$$

□

Corollary 1 (Survival probability)**.**

$$1 - E(\tau) = \frac{(1-\varepsilon)\,\psi(\tau)}{\psi(\tau) - \varepsilon}\,. \tag{7}$$

Proof. Direct computation: $1 - E = 1 - \frac{\varepsilon(1-\psi)}{\varepsilon-\psi} = \frac{(\varepsilon-\psi)-\varepsilon(1-\psi)}{\varepsilon-\psi} = \frac{\psi(\varepsilon-1)}{\varepsilon-\psi} = \frac{(1-\varepsilon)\,\psi}{\psi-\varepsilon}$. □

3.3 Branch Propagation Factor

Proposition 2 (TDBD branch factor)**.** *Along a branch from child age* τ_c *to parent age* τ_p*, the D-propagation factor is:*

$$\boxed{\frac{D(\tau_p)}{D(\tau_c)} = \frac{\psi_p\,(\psi_c - \varepsilon)^2}{\psi_c\,(\psi_p - \varepsilon)^2}}\,, \tag{8}$$

where $\psi_c = \psi(\tau_c)$ *and* $\psi_p = \psi(\tau_p)$*.*

Proof. From (4), the propagation factor is $\exp\!\big(-\int_{\tau_c}^{\tau_p} \alpha(s)\,\mathrm{d}s\big)$ where $\alpha = \lambda(1 + \varepsilon - 2E)$.

Step 1: Simplify $1 + \varepsilon - 2E$.

$$1 + \varepsilon - 2E = \frac{(1+\varepsilon)(\varepsilon-\psi) - 2\varepsilon(1-\psi)}{\varepsilon - \psi} = \frac{(1-\varepsilon)(\varepsilon+\psi)}{\psi - \varepsilon}\,.$$

Step 2: Change of variables. Since $\mathrm{d}\psi = (1-\varepsilon)\,\lambda(s)\,\psi\,\mathrm{d}s$, we have $\lambda(s)\,\mathrm{d}s = \mathrm{d}\psi/\big((1-\varepsilon)\,\psi\big)$.

$$\int_{\tau_c}^{\tau_p} \lambda(s)\,\frac{(1-\varepsilon)(\varepsilon+\psi)}{\psi-\varepsilon}\,\mathrm{d}s = \int_{\psi_c}^{\psi_p} \frac{\varepsilon+\psi}{\psi(\psi-\varepsilon)}\,\mathrm{d}\psi. \tag{9}$$

Step 3: Partial fractions.

$$\frac{\varepsilon+\psi}{\psi(\psi-\varepsilon)} = \frac{-1}{\psi} + \frac{2}{\psi-\varepsilon}\,.$$

Step 4: Integrate.

$$\int_{\psi_c}^{\psi_p} \left(\frac{-1}{\psi} + \frac{2}{\psi-\varepsilon}\right)\mathrm{d}\psi = \Big[-\ln\psi + 2\ln|\psi-\varepsilon|\Big]_{\psi_c}^{\psi_p} = \ln\frac{(\psi_p-\varepsilon)^2}{\psi_p} - \ln\frac{(\psi_c-\varepsilon)^2}{\psi_c}\,.$$

Taking the negative exponential:

$$\frac{D(\tau_p)}{D(\tau_c)} = \exp\!\left(-\int \alpha\,\mathrm{d}s\right) = \frac{\psi_p\,(\psi_c-\varepsilon)^2}{\psi_c\,(\psi_p-\varepsilon)^2}\,. \qquad \square$$

In logarithmic form (as implemented):

$$\log\frac{D(\tau_p)}{D(\tau_c)} = (1-\varepsilon)\big(\Lambda_p - \Lambda_c\big) + 2\log|\psi_c-\varepsilon| - 2\log|\psi_p-\varepsilon|, \tag{10}$$

where $\Lambda_c = \Lambda(\tau_c)$, $\Lambda_p = \Lambda(\tau_p)$.

3.4 Full TDBD Log-Likelihood

Assembling (1) using the branch factors and speciation-event contributions:

$$\log\mathcal{L}_{\text{TDBD}} = \log\Gamma(n) + \sum_{k=1}^{n-1}\log\lambda(\tau_k) + \sum_{\text{branches}}\log\frac{D(\tau_p)}{D(\tau_c)} - 2\log\!\big(1-E(T)\big), \tag{11}$$

where τ_k are the ages of the $n-1$ internal nodes (speciation events), and each branch factor is given by (10).

4 CRBD as a Special Case ($z = 0$)

When $z = 0$, rates are constant: $\lambda(\tau) = \lambda$, $\mu(\tau) = \mu$, $\varepsilon = \mu/\lambda$. Then $\Lambda(\tau) = \lambda\tau$ and $\psi(\tau) = \mathrm{e}^{(\lambda-\mu)\tau} = \mathrm{e}^{r\tau}$ where $r = \lambda - \mu$.

4.1 CRBD Extinction

Substituting into (6):

$$E(\tau) = \frac{\mu\,(\mathrm{e}^{r\tau}-1)}{\lambda\,\mathrm{e}^{r\tau}-\mu}\,. \tag{12}$$

4.2 CRBD Branch Factor

Define $g(\tau) = \log|\lambda\mathrm{e}^{r\tau}-\mu|$. From (10):

$$\log\frac{D(\tau_p)}{D(\tau_c)} = r\,(\tau_p-\tau_c) + 2\,g(\tau_c) - 2\,g(\tau_p). \tag{13}$$

4.3 CRBD Log-Likelihood

Since λ is constant, $\sum_k \log \lambda(\tau_k) = (n-1)\log \lambda$:

$$\log \mathcal{L}_{\mathrm{CRBD}} = \log \Gamma(n) + (n-1)\log \lambda + \sum_{\text{branches}} \Big[r\,(\tau_p - \tau_c) + 2g(\tau_c) - 2g(\tau_p) \Big] - 2\log\big(1 - E(T)\big). \quad (14)$$

4.4 Critical Case $\lambda = \mu$

When $r \to 0$, L'Hôpital's rule gives:

$$E(\tau) = \frac{\lambda\tau}{1+\lambda\tau}, \qquad 1 - E(\tau) = \frac{1}{1+\lambda\tau}, \qquad \log \frac{D(\tau_p)}{D(\tau_c)} = 2\log\frac{1+\lambda\tau_c}{1+\lambda\tau_p}. \quad (15)$$

5 CRB and TDB: Pure-Birth Models ($\mu = 0$)

5.1 CRB (λ constant, $\mu = 0$)

With $\varepsilon = 0$: $E(\tau) = 0$, $D(\tau)$ decays at rate λ, and no root conditioning is needed:

$$\log \mathcal{L}_{\mathrm{CRB}} = \log \Gamma(n) + (n-1)\log \lambda - \lambda\, S, \quad (16)$$

where $S = \sum_{\text{branches}} \ell_i$ is the total branch length. This matches the well-known Yule-process likelihood.

5.2 TDB ($\lambda(\tau)$ time-varying, $\mu = 0$)

With $\varepsilon = 0$: $E = 0$, and the branch factor simplifies to $\exp\big(-\int_{\tau_c}^{\tau_p} \lambda(s)\,\mathrm{d}s\big)$:

$$\log \mathcal{L}_{\mathrm{TDB}} = \log \Gamma(n) + \sum_{k=1}^{n-1} \log \lambda(\tau_k) - \sum_{\text{branches}} \Lambda(\tau_p, \tau_c), \quad (17)$$

where $\Lambda(\tau_p, \tau_c) = \int_{\tau_c}^{\tau_p} \lambda(s)\,\mathrm{d}s$ has the analytical expression in (5).

6 Degeneracy Relations

The model family has the following degeneracy hierarchy (Table 1):

Specialization	**Reduces to**	**Mechanism**
$\mathrm{CRBD}(\lambda, \mu{=}0)$	$\mathrm{CRB}(\lambda)$	$\varepsilon = 0$, no extinction
$\mathrm{TDBD}(\lambda_0, z, \varepsilon{=}0)$	$\mathrm{TDB}(\lambda_0, z)$	$\psi \to \mathrm{e}^{\Lambda}$, $E = 0$
$\mathrm{TDB}(\lambda_0, z{=}0)$	$\mathrm{CRB}(\lambda_0)$	$\lambda(\tau) = \lambda_0$
$\mathrm{TDBD}(\lambda_0, z{=}0, \varepsilon)$	$\mathrm{CRBD}(\lambda_0, \varepsilon\lambda_0)$	$\Lambda = \lambda_0 \tau$
$\mathrm{BAMM}(\eta{=}0, \mathrm{bg}, \mathrm{fg})$	bg	no switching
$\mathrm{BAMM}(\eta, M, M)$	M	identical regimes

Table 1: Degeneracy relations among models. All are verified numerically.

7 BAMM: Hybrid Analytical–Numerical Computation

7.1 Multi-State ODE System

A single-layer BAMM(η, bg, fg) model has two states: background (state 0) and foreground (state 1). The transition matrix is:

$$Q = \begin{pmatrix} -\eta & \eta \\ 0 & 0 \end{pmatrix}.$$

The extinction ODE becomes a coupled 2D system:

$$\frac{\mathrm{d}E_0}{\mathrm{d}\tau} = \mu_b - (\lambda_b + \mu_b + \eta)\, E_0 + \lambda_b\, E_0^2 + \eta\, E_1, \tag{18}$$

$$\frac{\mathrm{d}E_1}{\mathrm{d}\tau} = \mu_f - (\lambda_f + \mu_f)\, E_1 + \lambda_f\, E_1^2. \tag{19}$$

7.2 Key Observation: One-Way Coupling

Equation (19) for the foreground state is **independent** of E_0. Since $Q_{10} = 0$ (no reverse switching), the fg state is an *absorbing state*. Its equation is a standard single-state Riccati equation with closed-form solution (Section 3).

The background equation (18) depends on $E_1(\tau)$ through the forcing term $\eta\, E_1$. Even with E_1 known analytically, the resulting non-autonomous Riccati equation for E_0 does not generally admit a closed form, and must be solved numerically.

7.3 Dimension Reduction

Our hybrid strategy is:

1. **Identify absorbing states**: state i with $\sum_{j \neq i} |Q_{ij}| = 0$.
2. **Solve absorbing states analytically**: use the CRBD/TDBD closed form from Section 3.
3. **Reduce ODE dimension**: integrate only the non-absorbing states numerically, plugging in the analytical $E_j(\tau)$ for absorbing states j as known forcing functions.

For a single-layer BAMM with non-BAMM submodels, this reduces the extinction ODE from 2D to **1D**, and similarly for the D propagation.

7.4 Nested BAMM

For nested BAMM (e.g., BAMM(η_1, BAMM(η_2, M_1, M_2), M_3)), the state space grows:

$$Q = \begin{pmatrix} -\eta_2 - \eta_1 & \eta_2 & \eta_1 \\ 0 & -\eta_1 & \eta_1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Only state 2 (the ultimate fg) is absorbing. States 0 and 1 require numerical ODE integration, but the system is reduced from 3D to 2D.

In general, for an N-state system with k absorbing states, the ODE dimension is reduced from N to $N - k$.

7.5 D Propagation

The same decomposition applies to the D-propagation ODE. For absorbing state j with CRBD parameters (λ_j, μ_j):

$$D_j(\tau) = D_j(\tau_c) \cdot \frac{\psi_j(\tau)\,\big(\psi_j(\tau_c) - \varepsilon_j\big)^2}{\psi_j(\tau_c)\,\big(\psi_j(\tau) - \varepsilon_j\big)^2}, \qquad \tau \in [\tau_c, \tau_p].$$

For non-absorbing state i, the ODE is:

$$\frac{\mathrm{d}D_i}{\mathrm{d}\tau} = -\big(\lambda_i + \mu_i - Q_{ii} - 2\lambda_i E_i\big)\, D_i + \sum_{j \neq i} Q_{ij}\, D_j(\tau),$$

where $D_j(\tau)$ for absorbing j is evaluated analytically at each ODE step.

8 Numerical Stability

The closed-form expressions involve exponentials that can overflow. We employ the following safeguards:

1. **Log-space computation**: The branch factor (10) is computed entirely in log-space.
2. **Stable** $\log|\psi - \varepsilon|$: When $\log\psi > 50$, we use:

   $$\log|\psi - \varepsilon| = \log\psi + \log\!\big(1 - \varepsilon\, \mathrm{e}^{-\log\psi}\big),$$

   avoiding materialization of ψ.
3. **Critical-case branches**: When $|r| < 10^{-12}$ (CRBD) or $|1 - \varepsilon| < 10^{-12}$ (TDBD), we switch to the L'Hôpital limit formulas.
4. **Asymptotic limits**: When $r\tau \to +\infty$: $E \to \varepsilon$, $1 - E \to 1 - \varepsilon$. When $r\tau \to -\infty$: $E \to 1$, $1 - E \to 0$.

9 Computational Complexity

Model	Previous	Now	Speedup
CRB	$O(n)$ closed	$O(n)$ closed	—
TDB	$O(n)$ closed	$O(n)$ closed	—
CRBD	ODE per branch	$O(n)$ closed	$\gg 1\times$
TDBD	ODE per branch	$O(n)$ closed	$\gg 1\times$
BAMM (1-layer)	N-dim ODE per branch	$(N-k)$-dim ODE	$\sim 2\times$
BAMM (nested)	N-dim ODE per branch	$(N-k)$-dim ODE	depends on k/N

Table 2: Computational complexity comparison. n = number of tips, N = number of states, k = number of absorbing states with closed form.

For CRBD and TDBD, the improvement is most dramatic: from $O(n)$ calls to a BDF ODE solver (each requiring multiple Newton iterations and Jacobian evaluations) to $O(n)$ evaluations of elementary functions (exp, log).

10 Verification

All closed-form solutions are verified against high-precision numerical ODE integration (`scipy.integrate.solv` BDF method, `rtol`$=10^{-12}$, `atol`$=10^{-14}$). The test suite contains 118 checks:

- CRBD $E(\tau)$: 5 parameter sets $\times$ 5 time points = 25 checks.
- CRBD $\log(1 - E)$: 4 parameter sets $\times$ 3 time points = 12 checks.
- CRBD branch factor: 4 parameter sets $\times$ 3 branch segments = 12 checks.
- CRBD full likelihood vs. rate system: 5 checks.
- CRBD $\to$ CRB degeneracy: 3 checks.
- TDBD $E(\tau)$: 5 parameter sets $\times$ 5 time points = 25 checks.
- TDBD branch factor: 4 parameter sets $\times$ 3 branch segments = 12 checks.

- TDBD full likelihood vs. rate system: 4 checks.
- TDBD $\to$ TDB degeneracy: 3 checks.
- TDBD($z = 0$) $\to$ CRBD degeneracy: 3 checks.
- BAMM degeneracies ($\eta = 0$, bg = fg): 2 checks.
- BAMM hybrid (1-layer, 3-layer nesting): 3 checks.
- Noise mixing ($\sigma = 0, 1, \infty$): 3 checks.
- BAMM irreversibility: 2 checks.
- Original regression suite: 4 checks.

All 118 tests pass with tolerance 10^{-6}.

11 Summary of Formulas

Quantity	**Formula**
$\psi(\tau)$	$\exp\big((1-\varepsilon)\,\Lambda(\tau)\big)$
$E(\tau)$	$\dfrac{\varepsilon\,(1-\psi)}{\varepsilon-\psi}$
$1 - E(\tau)$	$\dfrac{(1-\varepsilon)\,\psi}{\psi-\varepsilon}$
$\log\dfrac{D(\tau_p)}{D(\tau_c)}$	$(1-\varepsilon)(\Lambda_p-\Lambda_c) + 2\log\lvert\psi_c-\varepsilon\rvert - 2\log\lvert\psi_p-\varepsilon\rvert$
CRBD specialization ($z = 0$, $\Lambda = \lambda\tau$, $\psi = \mathrm{e}^{r\tau}$, $r = \lambda - \mu$, $\varepsilon = \mu/\lambda$):	
$E(\tau)$	$\dfrac{\mu(\mathrm{e}^{r\tau}-1)}{\lambda\mathrm{e}^{r\tau}-\mu}$
$\log\dfrac{D(\tau_p)}{D(\tau_c)}$	$r(\tau_p-\tau_c) + 2\log\lvert\lambda\mathrm{e}^{r\tau_c}-\mu\rvert - 2\log\lvert\lambda\mathrm{e}^{r\tau_p}-\mu\rvert$

Table 3: Summary of closed-form solutions.